SOFTWARE REQUIREMENTS SPECIFICATION

FITNESS TRACKING APP

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for the Fitness Tracking App. It describes the expected behaviour, interfaces, data, constraints, and acceptance criteria so that developers, testers, and project reviewers share a common understanding.

1.2 Scope

The Fitness Tracking App helps users record and review everyday fitness activities. Core features include account creation and login, profile settings, step and activity logging, workout recording, goal setting, progress summaries, reminders, and optional synchronization with supported device health data.

The application is intended for personal activity awareness and habit tracking. It is not a medical diagnostic tool and does not replace advice from a qualified health professional.

1.3 Intended Audience

- Students and project evaluators
- Software developers and UI designers
- Quality assurance / testing team
- Project supervisors and stakeholders

1.4 Definitions and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
UI	User Interface

Term	Meaning
API	Application Programming Interface
GPS	Global Positioning System
FR	Functional Requirement
NFR	Non-Functional Requirement
User	A person registered with the application

1.5 References

The structure is informed by common SRS practices and ISO/IEC/IEEE 29148:2018 requirements-engineering guidance. This document is a project-specific draft, not a claim of formal certification.

2. Overall Description

2.1 Product Perspective

The app is a mobile-first software product. It may use device sensors or approved health-platform APIs to obtain activity data, while allowing manual entry when automatic tracking is unavailable. A backend service may store account and synchronized activity data.

2.2 Product Functions

- Register, sign in, sign out, and manage a user profile.
- Record daily steps, active minutes, distance, and estimated calories where supported.
- Start, pause, resume, and finish a workout session.
- Create personal activity goals and view progress.
- Display daily, weekly, and monthly summaries.
- Schedule optional reminders and notifications.
- Export or delete personal activity data.

2.3 User Classes

User class	Description
Guest	Can view introductory information and create an account.
Registered user	Can track activity, manage goals, and review personal history.
Administrator / support	If enabled, manages service health and support requests; access to personal data must be limited.

2.4 Operating Environment

- Android or iOS smartphone with a supported operating-system version.
- Internet connection for account access, backup, and synchronization; basic manual logging may work offline.
- Optional access to motion sensors, location services, and device health platforms, subject to user permission.

2.5 Design and Implementation Constraints

- The app must request permissions before accessing sensitive device data.
- The interface should be usable on common phone screen sizes.

- The app must handle missing sensor data and network interruptions gracefully.
- Only data needed for stated features should be collected.

2.6 Assumptions and Dependencies

- Users have access to a compatible smartphone.
- Automatic tracking accuracy depends on device sensors, permissions, and supported integrations.
- Push notifications depend on operating-system settings and user consent.
- Third-party APIs may change or become unavailable.

3. System Features and Functional Requirements

Each requirement uses a unique ID so it can be tested and traced.

3.1 Account and Profile Management

- **FR-01:** The system shall allow a user to create an account using the supported registration method.
- **FR-02:** The system shall validate required fields and show a clear error for invalid input.
- **FR-03:** The system shall allow registered users to sign in and sign out securely.
- **FR-04:** The system shall allow users to view and update profile details and preferences.
- **FR-05:** The system shall provide a password/account recovery process where applicable.

3.2 Activity Tracking

- **FR-06:** The system shall display daily activity totals such as steps and active minutes when data is available.
- **FR-07:** The system shall allow manual entry or correction of supported activity records.
- **FR-08:** The system shall identify whether an activity record was manually entered or synchronized from a device.
- **FR-09:** The system shall show when activity data was last updated.
- **FR-10:** If permissions are denied or sensor data is unavailable, the system shall explain the limitation and offer manual logging where possible.

3.3 Workout Management

- **FR-11:** The system shall allow users to select a workout type, such as walking, running, cycling, or strength training.
- **FR-12:** The system shall allow a workout to be started, paused, resumed, and completed.
- **FR-13:** The system shall save workout duration and any supported measurements.
- **FR-14:** The system shall allow users to view previous workout sessions and their details.

3.4 Goals and Progress

- **FR-15:** The system shall allow users to set, edit, and remove daily or weekly activity goals.
- **FR-16:** The system shall show progress toward each active goal.
- **FR-17:** The system shall provide daily, weekly, and monthly activity summaries.
- **FR-18:** The system shall show simple charts or visual indicators for trends over time.

3.5 Notifications and Settings

- **FR-19:** The system shall allow users to enable, disable, and schedule activity reminders.
- **FR-20:** The system shall respect device notification permissions and quiet-time settings where available.

- **FR-21:** The system shall allow users to configure units and basic display preferences.

3.6 Data Management

- **FR-22:** The system shall synchronize supported data when a network connection is available and synchronization is enabled.
- **FR-23:** The system shall avoid creating duplicate activity records during synchronization.
- **FR-24:** The system shall allow users to request an export of their activity data in a documented format.
- **FR-25:** The system shall allow users to request deletion of their account and associated personal data, subject to clearly disclosed retention requirements.

4. External Interface Requirements

4.1 User Interface

- The main dashboard shall show daily activity, goal progress, and a shortcut to start a workout.
- Navigation shall provide access to Dashboard, Workouts, History, Goals, and Profile/Settings.
- Forms shall provide labels, validation messages, and clear save/cancel actions.
- Text, contrast, and touch targets should support comfortable use on mobile screens.

4.2 Hardware Interfaces

The app may interface with the phone's motion sensors, GPS receiver, and wearable devices where supported. Hardware access must be permission-controlled. The app shall not imply that unavailable measurements are exact.

4.3 Software Interfaces

- Mobile operating-system APIs for notifications and sensor access.
- Optional health-platform integrations, such as platform-provided health data APIs, subject to user authorization.
- Backend API for account, activity, goal, and synchronization operations.
- Database service for persistent storage.

4.4 Communication Interfaces

Client-server communication shall use encrypted transport (HTTPS/TLS). The client shall show an understandable status when network operations fail and should retry transient synchronization failures safely.

5. Non-Functional Requirements

5.1 Performance

- **NFR-01:** On a typical supported device, the dashboard should load within 3 seconds when local data is available.
- **NFR-02:** Common user actions should provide visible feedback within 2 seconds under normal network conditions.
- **NFR-03:** Synchronization shall not block the user from viewing already saved local data.

5.2 Security and Privacy

- **NFR-04:** The system shall protect data in transit using current secure transport protocols.
- **NFR-05:** Authentication secrets shall not be stored in plain text.
- **NFR-06:** The system shall enforce access controls so users can access only their own personal records.

- **NFR-07:** The system shall explain why permissions are requested and shall provide settings to revoke optional integrations.
- **NFR-08:** The system shall collect only data required for enabled features and provide a privacy notice.

5.3 Reliability and Availability

- **NFR-09:** The app shall preserve saved activity records during temporary network outages.
- **NFR-10:** The app shall provide a clear message when an operation fails and shall avoid silently losing user-entered data.
- **NFR-11:** Synchronization operations shall be safe to retry without unintended duplicate records.

5.4 Usability and Accessibility

- **NFR-12:** Common tasks such as viewing daily progress or starting a workout should require a small number of clearly labelled steps.
- **NFR-13:** The interface should support readable text, adequate contrast, and screen-reader labels where the platform permits.
- **NFR-14:** Errors shall describe the problem and a reasonable next action in plain language.

5.5 Maintainability and Portability

- **NFR-15:** Requirements and major modules shall be documented to support future maintenance.
- **NFR-16:** The app shall be tested on the agreed list of supported OS versions and device sizes.
- **NFR-17:** Logs shall avoid storing passwords, tokens, or unnecessary personal activity details.

6. Data Requirements

6.1 Main Data Entities

Entity	Example attributes	Purpose
User	user_id, email/username, preferences, created_at	Account and settings
ActivityRecord	record_id, user_id, date_time, activity_type, source, steps, duration	Stores activity measurements
WorkoutSession	session_id, user_id, type, start_time, end_time, duration, notes	Stores completed workouts
Goal	goal_id, user_id, goal_type, target_value, period, active_status	Stores user-defined goals
Reminder	reminder_id, user_id, schedule, enabled	Stores reminder preferences

6.2 Data Validation

- Required fields shall be checked before saving.
- Dates, durations, and numeric measurements shall be validated for sensible ranges.
- Records shall be associated with the correct user account.
- The system shall identify units and data sources for measurements.

6.3 Retention and Deletion

Retention periods shall be documented by the project owner. Users should be able to export or delete their data. Any legally or technically required retention exception must be disclosed clearly.

7. Use Cases

UC-01: View Daily Activity

Actor: Registered user
Precondition: The user is signed in.
Main flow: The user opens the dashboard; the system loads available activity records; the system displays totals and progress toward goals.
Alternative flow: If no data is available, the system displays an empty state and explains how to add or sync activity.
Postcondition: Daily activity information is displayed.

UC-02: Record a Workout

Actor: Registered user
Precondition: The app is open and the user can access workout features.
Main flow: The user selects a workout type; starts the session; pauses or resumes if needed; ends the session; reviews and saves the summary.
Alternative flow: If the app is interrupted, it should recover or clearly indicate whether the session was saved.
Postcondition: The workout is stored in history.

UC-03: Set an Activity Goal

Actor: Registered user
Precondition: The user is signed in.
Main flow: The user opens Goals; selects a goal type and target; saves it; the system displays progress as activity data arrives.
Alternative flow: Invalid targets are rejected with an explanation.
Postcondition: The active goal is saved.

UC-04: Export or Delete Data

Actor: Registered user
Precondition: The user is signed in and re-authenticates if required.
Main flow: The user selects export or deletion; the system explains the effect; the user confirms; the system processes the request and reports status.
Postcondition: Data is exported or the deletion request is recorded and processed.

8. Constraints, Assumptions and Dependencies

- Activity measurements may be estimates and vary by device and data source.
- Location-based workout tracking requires user permission and suitable GPS availability.
- Some integrations require a compatible device, operating-system service, or account.
- Network-dependent features may be unavailable offline.
- The project team must choose the final technology stack, database, hosting, and authentication provider.

9. Verification and Acceptance Criteria

ID / area	Verification method	Acceptance criterion
FR-01 to FR-05	Functional tests	User can register/sign in, validation works, and profile changes persist.
FR-06 to FR-10	Functional + device tests	Activity data is displayed with source/update status; unavailable data is handled clearly.
FR-11 to FR-14	Workflow tests	Workout can be started, paused, resumed, completed, and found in history.
FR-15 to FR-18	Functional tests	Goals save correctly and summaries reflect stored records.

ID / area	Verification method	Acceptance criterion
FR-19 to FR-21	Settings tests	Reminders and preferences follow user selections and permissions.
FR-22 to FR-25	Integration + privacy tests	Sync avoids duplicates; export and deletion requests behave as specified.
NFR-01 to NFR-03	Performance tests	Target response times are met on agreed test devices and network conditions.
NFR-04 to NFR-08	Security review	Transport, authentication, access controls, permissions, and privacy notice are checked.
NFR-09 to NFR-17	Reliability/usability tests	Offline handling, error messages, accessibility checks, and maintenance documentation are reviewed.

The project is ready for academic acceptance when all mandatory requirements pass their agreed tests, known limitations are documented, and no critical data-loss or unauthorized-access defect remains.

10. Future Enhancements

- Wearable-device synchronization and richer health-platform integration.
- Optional friend groups or challenges with explicit privacy controls.
- Personalized activity insights and trend summaries.
- Additional language support and improved accessibility options.
- Web dashboard for viewing exported or synchronized activity history.

Appendix A: Glossary

Term	Definition
Activity record	A saved measurement or summary of a physical activity.
Workout session	A time-bounded exercise event recorded by the user or device.
Synchronization	The process of transferring supported data between the device and backend.
Offline mode	Use of locally available features when the internet connection is unavailable.

References

- ISO/IEC/IEEE 29148:2018, Systems and software engineering — Life cycle processes — Requirements engineering.
- IEEE SRS template examples and common software requirements specification practices.

Public reference: ISO/IEC/IEEE 29148:2018 overview.